# GABRIEL S. STUART

**Automation, Computer Vision & Systems Simulation Developer**

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Builds end-to-end multiphysics digital twins, industrial process simulators, and automated computer-vision pipelines in Python, C/C++, and Java — coupling first-principles thermodynamic/electrochemical models with cognitive multi-agent systems, discrete-event TSN networks, and statistical validation frameworks tested in clinical-trial data extraction.

## KEY TECHNICAL SKILLS

**Languages & Simulation:** Python (3.11+, Numba JIT, Cython), C/C++, Java 17; PEM/SOEC electrolysis & ATR thermodynamics, CoolProp EOS, PEMFC electrochemistry, event-driven & NER engines, gRPC co-simulation

**Networks & Multi-Agent Systems:** OMNeT++ 6.4 / INET 4.7 (IEEE 802.1Qbv TSN, Scheduled MAC), JaCaMo/Jason (BDI), PROSA/ADACOR, Contract-Net, 2PC, Cooperative A* AMR routing

**Vision, Validation & DevOps:** OpenCV, YOLO (Detection/Pose), ONNX, PaddleOCR, PROSAC/RANSAC, Lin's CCC, Cohen's Kappa, ECDF, CAPEX/OPEX & LCOH, Pytest, Hypothesis, GitHub Actions, Docker, PyQt6, PCB design

## PROFESSIONAL EXPERIENCE

**External Developer (Remote Contractor) — ROGER Renewable Energy** **11/2025 – 05/2026**

- Developed a dynamic physical simulation proof-of-concept (PoC) for a digital green hydrogen plant balancing PEM and SOEC electrolyzer production under variable load profiles.
- Modeled byproduct oxygen ($O_2$) recovery from PEM/SOEC stacks to partially fuel an Autothermal Reforming (ATR) unit, coupling mass/energy balances and leading CAPEX/OPEX & LCOH validation.

**Hardware & Robotics Developer (Builder) — Robota Competition Team (UFSC)** **07/2022 – 09/2023**

- Designed custom printed circuit boards (PCBs) integrating reflectance sensors, power regulation, and motor drivers, and planned line-follower control trajectories for competitive robotics tournaments.

## FEATURED ENGINEERING PROJECTS

**Plot-in & Generator — Clinical Chart Extraction & Synthetic Vision** Plot-in · Generator · **Python, ONNX, PyQt6**

- Built a vision pipeline (Plot-in) extracting data from PDFs/images across 8 chart types via YOLO detection/pose, PaddleOCR, and PROSAC/RANSAC subpixel calibration; in validation with pharmaceutical professionals extracting clinical-trial data.
- Created a 3-stage synthetic chart generator (Generator) inspecting Matplotlib Agg Artists in memory for pixel-exact boxes and YOLO-pose keypoints; backed by 290+ Pytest/Hypothesis tests, Lin's CCC & Cohen's Kappa validation, and CI/CD installers.

**h2_plant — Modular Dual-Path Hydrogen Plant Simulator** github.com/utczbr/h2_plant · **Python, Numba, PyQt6**

- Engineered a 6-layer event-driven simulator covering PEM/alkaline electrolysis and ATR routes, oxygen buffers, BESS, water treatment, compression, and gas mixing with a YAML topology builder and PyQt6 visual node editor.
- Accelerated thermodynamic evaluations via lookup tables (`LUTManager`) and Numba JIT compilation, integrating checkpointed `FlowTracker` conservation checks, OPEX/LCOH techno-economic modeling, and 95%+ Pytest coverage.

**Matrix_Factory & Matrix-Network — PEMFC & TSN Digital Twin** Matrix_Factory · Matrix-Network · **Python, Java, C++**

- Coupled a Numba/Cython Python physics engine with Java 17 (JaCaMo/Jason) PROSA/ADACOR holonic agents over gRPC in Matrix_Factory, modeling 5 PEMFC stations (Kamal-Sourour cure, slot-die coating, 316L stamping wear, VDI 2230 torque, 1D stack polarization) with Contract-Net, Cooperative A* AMR routing, and 30 parallel JVM Monte Carlo instances.
- Built custom C++ OMNeT++ 6.4 / INET 4.7 modules (`ScheduledDcf` MAC, `DeadlineSink`) in Matrix-Network and ran a 10,560-run full-factorial campaign across FIFO, 802.1Q, and 802.1Qbv TSN backbones and scheduled wireless access (850 µs State / 560 µs Order slots, 90 ms cycle) to export empirical ECDF latency/loss lookup tables.

## EDUCATION & LANGUAGES

**UFSC (Federal University of Santa Catarina) — B.S. in Control and Automation Engineering** **Expected 2027**

**Graduate Coursework (PPGEAS/UFSC, 2024–2026):** Completed 7 Master's/PhD-level courses as an undergraduate special student: *AI, Concurrency & Real-Time Systems, Renewable Energy Systems, Machine Learning, Mobile Robotics, Linear Dynamic Systems,* and *Multiagent Systems*.

**Languages:** Portuguese (Native) · English (Advanced)